

# HW Solutions

ECE 310: Digital Signals Processing, Spring 2025

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## Solution:

- (a) The filter is (i) Type-I GLP. Since  $\angle H_d(\omega)$  has only  $\pi$  or  $2\pi$  jumps and otherwise keeps constant slopes from  $-\pi$  to  $\pi$ , this is a GLP filter. As  $\angle H_d(0) = 0$ , the GLP is type-I or type-II. From (b), we know that the length is  $N = 7$ , which is odd. We can get the same conclusion that the filter is Type-I.
- (b) We know that  $-\alpha$  (the slope of the phase) and  $N$  (the length of the filter) are related through

$$\alpha = \frac{N-1}{2}$$

Since the slope is  $-3 = -\alpha$ , we can obtain  $\alpha = 3$  and  $N = 7$ .

- (c) Since the filter is Type-I with an odd length, it could be any type of filter, so (i), (ii), and (iii) are all correct.
- (d) As  $\angle H_d(\omega)$  jumps by  $\pi$  at  $\omega = \frac{\pi}{2}$ , it must correspond to a zero-crossing of  $R(\omega)$ . Therefore,  $H_d(\frac{\pi}{2}) = 0$

**Grading:** 18 points

- (a) 5 points
- (b) 5 points
  - +3 if correct relation is shown
  - +1 for correct  $N$
- (c) 3 points
  - 0 if (iv) is chosen
  - +n if n correct answer is chosen
- (d) 5 points

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## Solution:

(a) for symmetric  $h[n]$  with  $N$  even,

$$R(\omega) = 2 \sum_{n=0}^{\frac{N}{2}-1} h[n] \cos\left(\omega \left(\frac{N-1}{2} - n\right)\right)$$

**Solution** Because  $N$  is even, we can rewrite  $H_d(w)$  as

$$\begin{aligned} H_d(w) &= \sum_{n=0}^{N/2-1} h[n]e^{-jwn} + \sum_{n=N/2-1}^{N-1} h[n]e^{-jwn} \\ &= e^{-jwM} \left[ \sum_{n=0}^{N/2-1} h[n]e^{-jw(n-M)} + \sum_{n=N/2-1}^{N-1} h[n]e^{-jw(n-M)} \right] \end{aligned}$$

Letting  $k = N - 1 - n$ , the equation can be written as:

$$H_d(w) = e^{-jwM} \left[ \sum_{n=0}^{N/2-1} h[n]e^{-jw(n-M)} + \sum_{k=N/2-1}^0 h[N-1-k]e^{-jw(N-1-k-M)} \right]$$

if we set  $M = \frac{N-1}{2}$ , the equation above becomes:

$$H_d(w) = e^{-jwM} \left[ \sum_{n=0}^{N/2-1} h[n]e^{-jw(n-M)} + \sum_{k=N/2-1}^0 h[N-1-k]e^{-jw(M-k)} \right]$$

Since  $h[n]$  is symmetric,  $h[n] = h[N-1-n]$ . This makes the equation above

$$\begin{aligned} H_d(w) &= e^{-jwM} \sum_{n=0}^{N/2-1} (h[n]e^{-jw(n-M)} + h[n]e^{-jw(M-n)}) \\ &= 2e^{-jwM} \sum_{n=0}^{N/2-1} h[n] \cos(w(M-n)) \\ R(w) &= 2 \sum_{n=0}^{N/2-1} h[n] \cos\left(w\left(\frac{N-1}{2} - n\right)\right) \end{aligned}$$

(b) for symmetric  $h[n]$  with  $N$  odd,

$$R(\omega) = h\left[\frac{N-1}{2}\right] + 2 \sum_{n=0}^{\frac{N-3}{2}} h[n] \cos\left(\omega\left(\frac{N-1}{2} - n\right)\right)$$

**Solution** Because  $N$  is odd, we can rewrite  $H_d(w)$  as

$$H_d(w) = e^{-jwM} [h[M] + \sum_{n=0}^{M-1} h[n]e^{-jw(n-M)} + \sum_{n=M+1}^{N-1} h[n]e^{-jw(n-M)}]$$

similar to a), by setting  $M = \frac{N-1}{2}$  and letting  $k = N-1-n$ , this can be written as

$$H_d(w) = e^{-jwM} [h[M] + \sum_{n=0}^{M-1} h[n]e^{-jw(n-M)} + h[N-1-n]e^{-jw(M-n)}]$$

Since  $h[n]$  is symmetric,  $h[n] = h[N-1-n]$ . This makes the equation above

$$H_d(w) = e^{-jwM} [h[M] + \sum_{n=0}^{M-1} (h[n]e^{-jw(n-M)} + h[n]e^{-jw(M-n)})]$$

$$R(w) = h\left[\frac{N-1}{2}\right] + 2 \sum_{n=0}^{\frac{N-3}{2}} h[n] \cos(w(\frac{N-1}{2} - n))$$

(c) for antisymmetric  $h[n]$  with  $N$  even,

$$R(\omega) = 2 \sum_{n=0}^{\frac{N}{2}-1} h[n] \sin\left(\omega\left(\frac{N-1}{2} - n\right)\right)$$

**Solution** Because  $N$  is even, following the same strategy in (a) we can rewrite  $H_d(w)$  as

$$H_d(w) = e^{-jwM} \sum_{n=0}^{N/2-1} (h[n]e^{-jw(n-M)} + h[N-1-n]e^{-jw(M-n)})$$

Antisymmetric even-length sequence means  $h[n] = -h[N-1-n]$ , which leads to

$$H_d(w) = e^{-jwM} \sum_{n=0}^{N/2-1} (h[n]e^{-jw(n-M)} - h[n]e^{-jw(M-n)})$$

$$= 2e^{-jw(M+\frac{\pi}{2})} \sum_{n=0}^{N/2-1} h[n] \sin(w(M-n))$$

$$R(w) = 2 \sum_{n=0}^{N/2-1} h[n] \sin(w(\frac{N-1}{2} - n))$$

(d) for antisymmetric  $h[n]$  with  $N$  odd,

$$R(\omega) = 2 \sum_{n=0}^{\frac{N-3}{2}} h[n] \sin \left( \omega \left( \frac{N-1}{2} - n \right) \right)$$

**Solution** Because  $N$  is odd, following the same strategy in (b) we can rewrite  $H_d(w)$  as

$$H_d(w) = e^{-jwM} [h[M] + \sum_{n=0}^{M-1} h[n]e^{-jw(n-M)} + h[N-1-n]e^{-jw(M-n)}]$$

Antisymmetric odd-length sequence means  $h[n] = -h[N-1-n]$  and  $h[\frac{N-1}{2}] = h[M] = 0$ , which leads to

$$\begin{aligned} H_d(w) &= e^{-jwM} \sum_{n=0}^{M-1} (h[n]e^{-jw(n-M)} - h[n]e^{-jw(M-n)}) \\ &= 2e^{-jw(M+\frac{\pi}{2})} \sum_{n=0}^{M-1} h[n] \sin(w(M-n)) \\ R(w) &= 2 \sum_{n=0}^{\frac{N-3}{2}} h[n] \sin(w(\frac{N-1}{2} - n)) \end{aligned}$$

**Grading:** 21 points

- +7 expansion into exponentials
- +7 correct application of symmetry
- +7 correct result

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### Solution:

The impulse response  $d[n]$  of an ideal low-pass filter with cutoff frequency  $\omega_c$  is

$$d[n] = \frac{\sin(\omega_c n)}{\pi n}$$

which is of infinite length. Next, in order to obtain a causal, GLP filter with finite length  $N$ , we shift  $d[n]$  by  $\frac{N-1}{2}$  to the right and only focus on the value with  $n = 0, 1, \dots, N-1$ , then we obtain

$$g[n] = d \left[ n - \frac{N-1}{2} \right] = \frac{\sin \left( \omega_c \left( n - \frac{N-1}{2} \right) \right)}{\pi \left( n - \frac{N-1}{2} \right)}, \quad n = 0, 1, \dots, N-1 \quad (1)$$

which has even symmetry. After that, we apply the window function  $w[n]$  on  $g[n]$  which gives

$$h[n] = w[n]g[n] = w[n] \cdot \frac{\sin \left( \omega_c \left( n - \frac{N-1}{2} \right) \right)}{\pi \left( n - \frac{N-1}{2} \right)}, \quad n = 0, 1, \dots, N-1 \quad (2)$$

as our filter coefficients.

(a) Rectangular window: By definition,

$$w[n] = 1, \quad n = 0, 1, \dots, N-1$$

then by plugging  $w[n]$  and  $\omega_c = \pi/3$  into Eq. (2) we get

$$h[n] = \frac{\sin\left(\frac{\pi}{3}\left(n - \frac{N-1}{2}\right)\right)}{\pi\left(n - \frac{N-1}{2}\right)}, \quad n = 0, 1, \dots, N-1$$

(b) Hamming window: By definition,

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right), \quad n = 0, 1, \dots, N-1$$

then by plugging  $w[n]$  and  $\omega_c = \pi/3$  into Eq. (2) we get

$$h[n] = \left(0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right)\right) \cdot \frac{\sin\left(\frac{\pi}{3}\left(n - \frac{N-1}{2}\right)\right)}{\pi\left(n - \frac{N-1}{2}\right)}, \quad n = 0, 1, \dots, N-1$$

(c) The main difference between these two filters are on the transition band and the ripple. The filter results from the rectangular window has a thinner transition band but a larger ripple, in contrast, the filter from the Hamming window has a wider transition band but a smaller ripple.

**Grading:** 26 points

- (a) 10 points
  - -3 if  $(N-1)/2$  shift is missing
- (b) 10 points
  - -3 if  $(N-1)/2$  shift is missing
  - -3 if the window expression is wrong
- (c) 6 points

## 4

### Solution:

One can notice that the frequency response  $H_{\text{HP}}(\omega)$  of an ideal high-pass filter (HPF) with cutoff frequency  $\omega_c$  can be converted from the frequency response  $H_{\text{LP}}(\omega)$  of an ideal low-pass filter (LPF) with cutoff frequency  $\pi - \omega_c$ , via shifting  $H_{\text{LP}}(\omega)$  to the right by  $\pi$ . That is

$$H_{\text{HP}}(\omega) = H_{\text{LP}}(\omega - \pi)$$

See Figure 1 for an illustration.

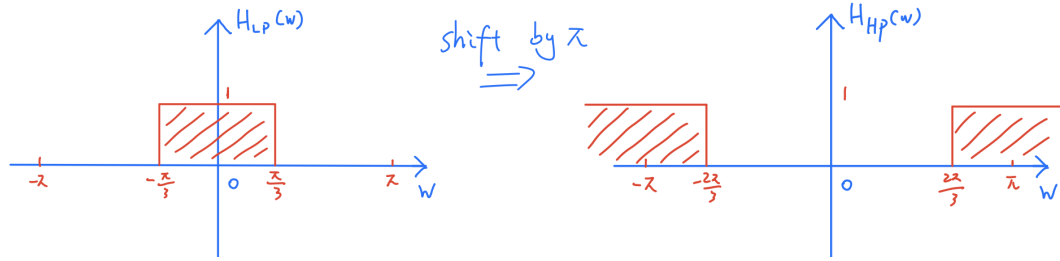


Figure 1: Transforming a low-pass filter to a high-pass filter.

Then, according to the frequency-shifting property of DTFT, let  $g_{LP}[n]$  and  $g_{HP}[n]$  be the (shifted) impulse responses of the ideal LPF and HPF, we have

$$g_{HP}[n] = e^{-j\pi n} g_{LP}[n] = (-1)^n g_{LP}[n]$$

In this way, by using the LPF impulse response in Eq. (1) from Problem 4 we get

$$\begin{aligned} g_{HP}[n] &= \frac{(-1)^n \sin\left((\pi - \omega_c)\left(n - \frac{N-1}{2}\right)\right)}{\pi\left(n - \frac{N-1}{2}\right)}, \quad n = 0, 1, \dots, N-1 \\ &= \frac{(-1)^n \sin\left(\frac{\pi}{3}\left(n - \frac{31}{2}\right)\right)}{\pi\left(n - \frac{31}{2}\right)}, \quad n = 0, 1, \dots, 31 \end{aligned}$$

with  $\omega_c = 2\pi/3$  and  $N = 32$ .

**Grading:** 20 points

- +10 if  $H(w)$  is characterized correctly
- +10 correct answer

## 5

### Solution:

**Relation between the frequency responses:** By definition, the window function  $w[n]$  is applied on the (shifted) impulse response  $g[n]$  of an ideal filter, and the resulting filter coefficient  $h[n]$  is

$$h[n] = g[n] \cdot w[n]. \quad (3)$$

Therefore, by applying the multiplication property of DTFT, the frequency response  $H_d(\omega)$  (the DTFT of  $h[n]$ ) satisfies

$$H_d(\omega) = \frac{1}{2\pi} \int_{2\pi} G_d(\theta) \cdot W_d(\omega - \theta) d\theta$$

where  $G_d(\cdot)$  and  $W_d(\cdot)$  are the DTFT of  $g[n]$  and  $w[n]$  respectively. Therefore, we can see that the frequency response  $H_d(\omega)$  is the (periodic) convolution between the frequency response of the ideal filter and the one of the window function.

**Why windowing does not destroy the GLP property:** By definition, a GLP filter requires its filter coefficients  $h[n]$  to have odd or even symmetry. From Eq. (3), notice that both the impulse response  $h[n]$  (e.g. the ideal LPF in Eq. (1)) and the windowing function (e.g. the rectangular and the hamming window) have either even or odd symmetry. Then, when such two symmetric sequences multiply together, the resulting sequence  $h[n] = g[n]w[n]$  should also have even or odd symmetry. Therefore, the windowing operation does not destroy the GLP property, as long as the window function is designed properly.

**Grading:** *15 points*

- +5 GLP requires symmetry
- +5 window function has symmetry
- +5 window does not affect symmetry